

6 In March 2019, a tropical cyclone in the south-west Indian Ocean caused widespread flooding in a number of countries.

The flooding and the damage caused by the cyclone meant that many people were at serious risk of cholera.

After the natural disaster occurred, many different areas reported outbreaks of cholera. Within a short time the disease had spread widely and large numbers of people were affected.

(a) Cholera is caused by a bacterial pathogen.

Name the bacterial pathogen that causes cholera.

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(b) Suggest **and** explain why the people affected by the cyclone were at serious risk of cholera.

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- (c) In addition to the standard treatment for cholera, antibiotics are recommended for people who are moderately ill or seriously ill with the disease.

- (i) Doxycycline is one of the main antibiotics used for the treatment of cholera.

Doxycycline enters the pathogen and binds to one of the subunits of the bacterial ribosome. This prevents growth and reproduction of the bacterial cell.

Suggest **and** explain how binding of doxycycline to ribosomes stops growth of the bacterial cell.

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- (ii) Penicillin, which used to be prescribed for the treatment of cholera, has a different mechanism of action to doxycycline.

State which part of the bacterial cell is affected by the action of penicillin.

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(d) Mozambique was one of the countries badly affected by the cyclone.

As part of the effort to prevent a greater number of cases of cholera from occurring, two different approaches were taken.

- Approximately 900 000 doses of the oral cholera vaccine were sent to Mozambique and a large-scale vaccination programme was organised.
- Medical centres were set up in Mozambique to treat people with cholera.

Suggest **and** explain how the **two** different approaches helped to prevent a greater number of cases of cholera from occurring.

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